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Analysis of pedestrians' behavior at pedestrian crossings

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Abstract

This paper presents a methodology for studying pedestrians' behavior at pedestrian crossings. To help understand the behavior of pedestrians, separate models were developed and estimated for divided and undivided streets. Estimated models include waiting time at the curbside and the number of crossing attempts needed by the pedestrian to make a successful crossing. From a broad range of road user and roadway factors, the strongest and most significant predictors which influenced the pedestrian's waiting time and the frequency of attempts to cross the streets were gender, age, number of children in household, crossing frequency, number of people in the group attempting to cross, access to private vehicle, destination, home location in relation to pedestrian crossing, and pedestrian past involvement in traffic accidents. In addition to these predictors, maximum likelihood estimates revealed that the pedestrian expected waiting time seems to profoundly influence the number of attempts needed to successfully cross the street (divided or undivided). Furthermore, results relating to pedestrian crossings on divided streets indicate that the pedestrian's expected waiting time when crossing from one side of the street to the central refuge island seem to increase the risk to end the waiting time when crossing from the refuge island to the other side of the street. Finally, results seem to suggest that pedestrians behave differently or have different waiting times as they cross from one side of the street to the refuge and from the refuge to the other side of the street. © 2001 Elsevier Science Ltd. All rights reserved.

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1. Introduction

Traffic accidents involving pedestrians have become a major safety problem all over the world, particularly in developing countries, due to high population density, rapid urbanization, and lack of adherence to traffic regulations by both drivers and pedestrians. Lack of adherence to traffic regulations at pedestrian crossings particularly by drivers create a paradigm in which pedestrians may become bold and force approaching vehicles in the traffic stream to brake in order to gain priority at the pedestrian crossing. On the other hand, pedestrian crossings with heavy pedestrian flow is likely to cause unacceptable vehicular delay (increased driver's frustration).

Studies concerning accidents involving pedestrians have addressed a number of essential issues. These issues include delay at pedestrian crossings (Griffiths et al., 1984), drivers' behavior and pedestrians interaction at pedestrian crossings (Katz et al., 1975; Griffiths et al., 1976; Fegan, 1978; Cresswell and Hunt, 1979; Landles, 1983; Himanen and Kulmala, 1988; Song et al., 1993; Varhelyi, 1998). A number of factors have been considered in these studies including: factors relating to physical environment (e.g. road width, existence of a central refuge at the crossing), road user variables (e.g. age, gender, socioeconomic characteristics, marital status), and the number of pedestrians in the group attempting to cross. Some of these factors turned out to influence the pedestrian's decision to cross the roadway.

Griffiths et al. (1984) developed models that describe delays at sites without a central refuge and with effectively a single stream of vehicles in each direction. The paper addressed the issue of delay at pedestrian crossings where the major source of delay is attributed to the interaction between vehicles and pedestrians. The study reported that most drivers were prepared to stop when a pedestrian occupied or was approaching their part of the roadway. Young children and elderly pedestrians were likely to step onto the roadway when it was clear that a driver had yielded right of way or if the road was clear of approaching vehicles. The study indicated that as vehicle delay increases, drivers become more aggressive, taking advantage of very small gaps, in pedestrian flow across the pedestrian crossing. Katz et al. (1975) reported that drivers slowed down or gave the right-of-way more often for crossing pedestrians as a group, rather than as an individual.

Recent empirical results by Varhelyi (1998) indicated that drivers' willingness to give the right of way to pedestrians at pedestrian crossings is low. Furthermore, drivers did not sufficiently lower their speeds to maintain a readiness to react to an unexpected dangerous situation. Moreover, drivers were willing to slow or stop when the speed of their vehicles was low. The study reported that three out of four drivers maintain the same speed and only one out of four slow down at crossing. The study concluded that drivers have to be influenced before they reached the decision zone (40–50 m before the pedestrian crossing) to prevent risky behavior.

Himanen and Kulmala (1988) specified and estimated a discrete choice model aiming to study the encounters between drivers and pedestrians at pedestrian crossings. The study addressed the problem by considering the choices of both drivers and pedestrians. Taking a variety of pedestrian's, environmental, and traffic conditions into consideration, the probabilities of a driver braking or weaving and of a

pedestrian continuing to cross in response to an encounter were identified. The data were collected through a video camera, as such, information relating to pedestrian's type of trip in pursuit and socioeconomic characteristics were not gathered. The study reported that safety margins exists when the speed of the approaching vehicle is low. For example, if the speed of the vehicle 20 m prior to the pedestrian crossing is 50 km/h, there is no safety margin in the encounter. The study indicated that the most important explanatory variables were: number of pedestrians in the group, city size, vehicle speed, vehicle platoon size, and pedestrian distance from curb.

Adherence to traffic regulations by both pedestrians and drivers in developing countries are low in comparison with that of developed nations. In Jordan, drivers' inclination to slow down and give way to pedestrians at pedestrian crossings is very low. Pedestrians usually wait at the curbside with the knowledge that drivers are not willing to stop and give them the right of way. As such, they try to force their way across hoping that drivers will slow or stop. If unsuccessful, they will return to the curbside and attempt again until successful. This crossing paradigm is likely to increase the pedestrians' waiting time and therefore the potential increases in the rate of pedestrian accidents. In making recommendations to reduce pedestrian accidents it is essential to study behavioral aspects of pedestrians at pedestrian crossings. Previous research has made theoretical and methodological contribution to a practical understanding of pedestrian's behavior and the interaction between the driver and the pedestrian at pedestrian crossings.

It is postulated in this paper that pedestrians arriving at the pedestrian crossing look for acceptable gaps between vehicles in the traffic stream. They either accept or reject such gaps. Rejection of prevailing gaps leads to longer waiting time at the curbside. The pedestrian's waiting time at the curbside is modeled in this paper by the risk function. If a pedestrian attempted to cross the road and was successful then this means that the waiting time at the curbside has ceased (the pedestrian has accepted the available gap between vehicles). If on the other hand the attempt was not successful; then the waiting time has not ceased and the pedestrian has to make more attempt(s) to cross the road. It is hypothesized in this paper that as the number of crossing attempts increase, pedestrians in turn are likely to be bold and thereby force approaching vehicles to brake (i.e. increase the risk of crossing). As such, the pedestrian's expected waiting time will enter the number of crossing attempts model as an exogenous variable.

In addition to the different approach adopted here, this paper will consider other predictors not considered in past research efforts. These predictors include origin and destination of the pedestrian, pedestrian's past involvement (directly involved or witnessed) in traffic accidents, marital status, accessibility to private vehicle, and number of children in the household. In addition, this study will develop separate models (duration and count) to model the behavior of pedestrians on both divided and undivided streets. Moreover, separate models for each direction of divided streets will be estimated. Although models in this paper are calibrated with data gathered in Jordan, the methodology outlined here can be applied in any setting. Indeed, earlier studies that addressed the behavior of pedestrians at pedestrian crossings concluded large differences in crossing behavior among different countries.

2. Model formulation

2.1. Pedestrians' waiting time at the curbside

The distribution of pedestrians' waiting time (t) at the pedestrian crossing, considered as a random variable, is characterize in this paper by the risk function $\xi(t)$. This function gives the instantaneous failure rate (ceasing the pedestrian's waiting time at the curbside) assuming that the pedestrian has not crossed successfully (i.e. rejected available gaps) the street to time t. The notion of the risk function is a convenient theoretical construct that provides an index of the relative levels of pedestrians' risk taking. Survival theory has been used extensively in the field of transport in the last 10 years (Jovanis and Chang, 1989; Hensher and Mannering, 1994; Paselk and Mannering, 1994). This can be attributed to: (1) survival theory provide some conceptual or theoretical ground to transport problems; (2) survival theory provide the researcher with flexibility with regards to the underlying distribution of the duration data; and (3) survival models can be estimated with ease due to the availability of statistical software.

The pedestrian's waiting time at the curbside is assumed to depend on exogenous variables. As such, the risk function can be stated as:

$$\xi(t|\mathbf{R}) = \lim_{\Delta t \downarrow 0} \Pr[t \leqslant T < t + \Delta t | T \geqslant t, \mathbf{R}]/\Delta t \tag{1}$$

where $\mathbf{R}$ is a column vector of exogenous variables. Since all pedestrians were observed to complete the duration of crossing, observations are said to be non-censored. The probability density function (Φ (t)) can be given as (Andersen, 1991):

$$\Phi(t|\mathbf{R}) = \xi(t|\mathbf{R})\exp[-\int_0^t \xi(s)ds] \tag{2}$$

One of the popular mathematical models used for analyzing waiting time data is the Cox proportional hazards model (Cox, 1972; Andersen, 1991). In the context of this model, the risk function at time t for pedestrian i can be written as:

$$\xi(t, \mathbf{R}_i) = \bar{\xi}(t) f(\mathbf{R}_i, \chi) \tag{3}$$

where $\bar{\xi}(t)$ is referred to as the underlying risk or hazard, (., .) is some non-negative function, and χ is a row vector of unknown parameters. In the context of pedestrians crossing streets, the $\mathbf{R}_i$ is time-independent. In most duration models, the predictors are assumed time-independent. A time-independent variable is defined as the variable whose values for a given pedestrian do not change during the spell (time under study). For example, marital status for an individual might change during the spell of unemployment. However, in our case, the issue of time-dependent variables is not critical since the waiting time on the pedestrian crossing is measured in seconds. To model the association between vector of predictors $\mathbf{R}$ and the waiting time t, the Cox proportional hazards model can be written as:

$$\xi(t|\mathbf{R}) = \bar{\xi}(t)e^{\sum_{i=1}^{K}\chi_i \mathbf{R}_i} \text{ with } 0 \leqslant \xi(t|\mathrm{R}) < \infty \tag{4}$$

where $\mathbf{R} = \{\mathrm{R}_1, \mathrm{R}_2, \mathrm{R}_3, \ldots, \mathrm{R}_\mathrm{K}\}$ and χ is a vector of estimable parameters. Clearly, the above model gives an expression for the risk at time t for a pedestrian with a given specification of a set of predictors ($\mathbf{R}$). The proportional hazard model is the product of the baseline risk function and the exponential expression "e" to the linear sum of $\chi_\mathrm{i}\mathbf{R}_\mathrm{i}$. The exponential expression describes the risk of failure for pedestrian with regression variable R relative to that at a standard value $\mathbf{R} = 0$ (at all time points). In the context of this model an independent variable with a negative estimated coefficient lowers pedestrians' risk of ceasing their waiting time at the curbside and thus increases the expected waiting time. While a positive coefficient increases pedestrians' risk of ceasing their waiting time and thus decrease the expected waiting time.

An important feature of this model is that the baseline risk is a function of time, but does not involve the **R**'s. In contrast, the exponential expression involves the **R**'s but does not involve t. If all the **R**'s are equal to zero, the model reduces to the baseline hazard function. Another important property of the proportional hazard model is that allowing $\bar{\xi}(t)$ to be quite general has the advantage of providing reasonably good relative efficiency for the estimation of **R** without having to make assumptions about $\bar{\xi}(t)$, thus making this model a semi-parametric duration model. In general, it is preferred to use a parametric model if we were sure of the correct model. Thus, when in doubt, the proportional hazard model is likely to give reliable results so that it is a safe choice of model (Cox, 1972; Andersen, 1991; Chap, 1997). Past research have demonstrated the robustness of the proportional hazard model. Other reason for making this model attractive is its form. The exponential part of this model ensures that the fitted model will always give estimated hazards that are non-negative.

2.1.1. Maximum likelihood estimates

Maximizing the likelihood function derives maximum likelihood estimates of the model's parameters. The maximization process is carried out by taking partial derivatives of the log-likelihood function with respect to each parameter in the model, and then solving a system of equations. Previous literature have shown that in the absence of censoring, which is the case in our problem, partial likelihood is precisely the likelihood based on the marginal of the ranks of the failure times (Cox, 1972). Observed pedestrians waiting times showed ties in the data and absence censoring. The partial log-likelihood (lnL) is defined as the sum of the pedestrians' likelihood's. Let C_i represent pedestrians ending their waiting time at the same time t_i. To account for ties in the data, the partial log-likelihood function can be written as:

$$\ln L = \sum_{i=l}^{N}[\chi' \sum_{j \in T_i} \mathbf{R}_j - C_i \ln \sum_{j \in E_j} e^{\chi' \mathbf{R}_j}] \tag{5}$$

where E_i represents the index set of pedestrians with waiting time greater than or equal to t_i. As such, for every pedestrian j in the set k_i, $t_j \geqslant T_i$, where the T's are the pedestrians' waiting times at the curbside. The partial log-likelihood is maximized using Newton's iterative method.

2.1.2. Goodness-of-fit

The likelihood ratio (LR) test is used to determine the overall goodness-of-fit of the developed pedestrian waiting time model. The likelihood can be defined as the probability of the observed data being explained by a particular model. The fit of the estimated model can be tested using the LR test. Formally, this test is specified as:

$$\mathrm{LR} = -2(\mathrm{LL}_z - \mathrm{LL}_c(K)) \tag{6}$$

where LL_z is the log-likelihood at zero with all the regression coefficients are set as zero and LL_c is the log-likelihood at convergence with K regression coefficients, respectively. The LR has a χ^2 distribution with K degrees of freedom.

2.2. Number of crossing attempts

Pedestrians at the pedestrian crossing usually make a number of attempts before they successfully cross the street. They carry out these attempts because drivers do not yield the right of way to pedestrians. Pedestrians will remain waiting at the curbside (except for risk takers) and make further attempts until successful. It is assumed that the number of crossing attempts, a random variable, is influenced by a complex interaction of variables. The objective is to model the number of attempts needed to successfully cross the street. In accounting for the randomness of traffic disruption, the obvious choice for the distribution of the number of crossing attempts (q) is the Poisson distribution with parameter δ. In many cases, the Poisson distribution is implausible, as it implies that the mean and the variance of q are equal. A convenient generalization is the assumption that δ_i for pedestrian i is random itself. A well-known and plausible specification is the assumption that δ is distributed per pedestrian according to a Γ-distribution. The resulting distribution of P is the negative binomial distribution (Gourieroux et al., 1984; Heckman and Singer, 1984). As such, the negative binomial model accounts explicitly for the dispersion in the data.

Observed number of crossing attempts made by pedestrians show that the mean and the variance are not exactly the same. As such, both the Poisson and the negative binomial models will be estimated. The distribution for the number of crossing attempts as well as the mean and the variance are shown in Fig. 1.

The Poisson probability P_r (q_i) of pedestrian i making q attempts to cross the street can be given as:

$$P_r(q_i) = \exp(-\delta_i)\delta_i^{q_i}/(q_i)! \tag{7}$$

where δ_i is the Poisson model parameter for pedestrian i which will be a function of measured external variables (vector $\Re$). Formally, this parameter can be written as:

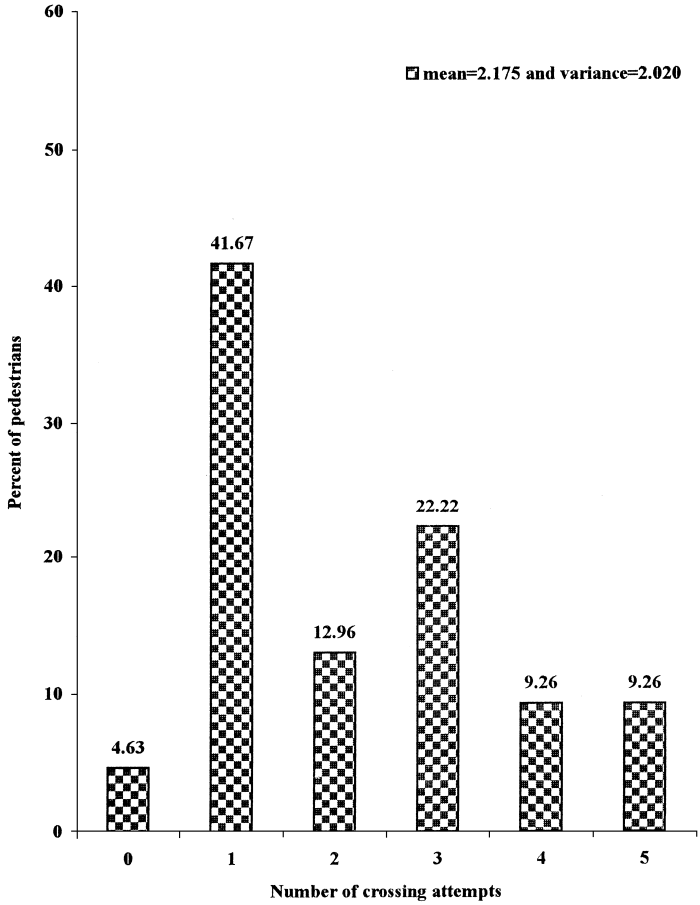

Fig. 1. Distribution of crossing attempts for undivided streets.

$$\ln(\delta_i) = \upsilon\Re_i \tag{8}$$

where υ is a vector of estimable parameters. These parameters are estimated from the likelihood function for N observed pedestrians. Formally, the likelihood and log-likelihood functions can be written respectively as:

$$L(\upsilon) = \prod_{i}^{N} \frac{\exp[-\exp(\upsilon\Re_i)][\exp(\upsilon\Re_i)]^{q_i}}{(q_i)!} \tag{9}$$

$$\ln L(\upsilon) = \sum_{i=l}^{N} [-\delta_i + q_i(\upsilon'\Re_i) - \ln(q_i!)] \tag{10}$$

and the partial derivatives are:

$$\partial \ln L(\upsilon)/\partial \upsilon = \sum_{i=l}^{N} [q_i - \delta_i] \Re_i = 0 \tag{11}$$

The negative binomial model is formulated by adding and error term ε to Poisson model parameter (Lawless, 1987). If the random variable exp (ε_i) is assumed to be gamma distributed with mean one and variance α^2, then the individual probability of having q number of attempts can be written as:

$$P_r(q_i|\Re_i) = \int_{-\infty}^{\infty} \frac{e^{-\exp(\upsilon'\Re_i+\varepsilon_i)} e^{(\upsilon'\Re_i+\varepsilon_i)q_i}}{(q_i)!} f(\varepsilon_i) d\varepsilon_i, \quad q_i = 0, 1, 2, \ldots \tag{12}$$

or

$$P_r(q_i|\Re_i) = \frac{\Gamma(q_i + h)[\alpha \exp(\upsilon'\Re_i)]^{q_i}}{\Gamma[h(q_i)!][1 + \alpha \exp(\upsilon'\Re_i)]^{h+q_i}} \tag{13}$$

where $h = 1/\alpha$, both α and υ are parameters estimated by maximum likelihood method. If the estimated α is significantly greater than zero we conclude that there is no equi-dispersion in the data and reject the assumption that q_i is Poisson distributed (Gourieroux et al., 1984). As such, we have $q \sim$NB (δ; α). In the limiting case $\alpha \rightarrow 0$ yields the Poisson distribution. Clark and Perry (1989) addressed the issue of estimating the dispersion parameter given a sample of independent, identically distributed values.

3. Data collection

In Jordan, the inclination of drivers to give way to pedestrians at pedestrian crossings is very low. According to 1997 figures, pedestrian accidents accounted for 15% of the total accidents in Jordan. In addition, 35% of total traffic-related injuries and 42% of traffic-related fatalities involved pedestrians. In the capital city of Amman, 54% of fatalities (all fatalities resulting from all types of accidents) are attributed to pedestrian accidents. Moreover, 52% of pedestrians' accidents causation is due to drivers not yielding the right of way to pedestrians.

Pedestrians attempting to cross pedestrian crossings were observed at 10 mid-block pedestrian crossings located in urban settings with level terrain in Amman. The first set of pedestrian crossings consists of five crossings located on undivided streets and the second set located on streets with a dividing central refuge. Each set has approximately the same geometry and traffic related characteristics. The crossings were selected provided that an approaching driver would be able to observe a pedestrian at the pedestrian crossing well in advance.

At pedestrian crossings, each pedestrian was monitored from the time he/she arrived at the crossing until he/she had successfully crossed the street. The arrival and the departure time (time the pedestrian stepped off the curb with the intention

of crossing) for each pedestrian were recorded. The waiting time for each pedestrian was taken as the difference between these two times. During this time the number of crossing attempts made by each pedestrian were recorded. Simultaneously, traffic volume, vehicle headway and approaching vehicle speed were recorded (40 m prior to the pedestrian crossing). The pedestrian monitoring process was discrete in order not to influence pedestrian behavior at the pedestrian crossing. Once the pedestrian had successfully crossed, he/she was asked a set of questions relating to marital status, age, number of children, availability of private vehicle, home location, origin and destination (e.g. home, work) of pursuit activity, and whether he/she had ever been involved in a pedestrian accident or had ever witnessed one.

Data was collected at the end of September 1998. In all, 400 pedestrians were monitored, however only 350 agreed to be interviewed. Information was gathered at different times of the day covering both a.m. and p.m. peak and off peak periods. Information relating to pedestrian crossing geometry included number of lanes in each direction and street width (curb to curb and from curb to central refuge island). It was noticed that at some pedestrian crossings pedestrians arrived in groups rather than singly. In addition, drivers did not lower their driving speeds significantly in order that they may be prepared to stop in an unexpected dangerous situation. Thus, giving total disregard to the law concerning giving the right of way to pedestrians.

4. Empirical results

4.1. Undivided streets

Table 1 shows maximum likelihood estimation results of the pedestrians waiting time at pedestrian crossings. The likelihood ratio test clearly demonstrates the overall goodness-of-fit of the estimated waiting time model. The statistical significance of each individual variable is given by the t-ratio statistic. Most of the included variables are statistically significantly different from zero at the 10% level (one-tailed t-test).

Estimation results show that pedestrians who were involved or witnessed pedestrian accident(s), have a lower risk of ceasing their waiting time at pedestrian crossings. In other words, these pedestrians have higher waiting time. In addition, female pedestrians, pedestrians with children, pedestrians who own and drive private vehicles, and older pedestrians are not likely to accept higher risk and cease their waiting time at the curbside. Griffiths et al. (1984) reported that elderly pedestrians were only prepared to step onto the roadway when it was clear that a driver had yielded the right of way or the road was clear of approaching vehicles. Pedestrians who own and drive private vehicles have a greater perception of risk than non-drivers. On the other hand, the waiting time at pedestrian crossings seem to decrease if the pedestrian is destined to work, frequently uses the pedestrian crossing, short vehicle headway, or the number of people in a group attempting to cross is high. These results support the notion that when pedestrians are in a hurry to get to the work

Table 1
Pedestrian's waiting time at pedestrian crossings on undivided streets[a]

Variable	Coefficient/t-ratio[a]
If the pedestrian was involved or have witnessed an accident (1 if yes, 0 otherwise)	−1.1640/−2.350
If the pedestrian owns and drives a car (1 if yes, 0 otherwise)	−0.877/−2.467
Number of people in the group attempting to cross	1.858/2.133
Destined to work (1 if yes, 0 otherwise)	0.612/1.600
Vehicle time headway (in seconds)	1.688/1.939
Number of children in household	−0.364/−1.823
Gender (1 if female)	−0.959/−3.071
Age in years	−0.184/−2.524
Crossing frequency (1 if frequently, 0 otherwise)	0.291/1.303
Number of observations	350
Log-likelihood at convergence model	−398.004
Chi-squared ($k=9$) model	64.089
Significance level model	0.0000
Log-rank test with 9 degrees of:	
Chi-squared value	49.089
Probability	0.0000

[a] Critical values for t-distribution: (1) ±1.645 at the 5% level one-tailed t-test; (2) ±1.282 at the 10% level one-tailed t-test.

place they are more likely to accept higher risk and cease their waiting time. Furthermore, pedestrians waiting time is likely to decrease with increased pedestrian flow as more pedestrians are able to take advantage of an established pedestrian precedence. Griffiths et al. (1984) and Himanen and Kulmala (1988) reported similar outcomes. Katz et al. (1975) indicated that drivers give the right of way more often for crossing pedestrians as a group, rather than as individuals.

To assess the effect of included variables on the waiting time at pedestrian crossings, the hazard ratios for both indicator and continuous variables are computed. Table 2 shows estimates of hazard ratios and their corresponding 95% confidence intervals for indicator variables for pedestrians crossing undivided streets. Estimates show that pedestrians who had never been involved or did not witness pedestrian accidents are 3.205 times more likely than those who have witnessed or were involved in pedestrian accidents to have shorter waiting times at pedestrian crossings. In addition, pedestrians who have no access to private cars are 2.404 times more likely than those who own and drive private cars to have shorter waiting times at pedestrian crossings.

Pedestrians who are destined to work are 1.84 times more likely than those who are destined to other places to have shorter waiting times. Male pedestrians are 2.611 times more likely than female pedestrians to have shorter waiting times. Finally, pedestrians who frequently use pedestrian crossings are 1.338 times more likely to have shorter waiting times at pedestrian crossings.

Figs. 2–4 display the hazard ratios for continuous variables; age, number of children, and vehicle headway respectively. The hazard ratios for these variables are

Table 2
Estimates of hazard ratios[a] for pedestrians crossing undivided streets

Indicator variable	Hazard ratio	95% Confidence interval
If pedestrian was not involved or did not witness a pedestrian accident	3.205	1.148–6.063
If a pedestrian has no access to a private car	2.404	0.995–4.042
If a pedestrian is destined to work	1.844	0.871–3.907
Gender (1 if male)	2.611	0.884–3.884
Crossing frequency (1 if frequently)	1.338	0.864–2.071

[a] For indicator variables.

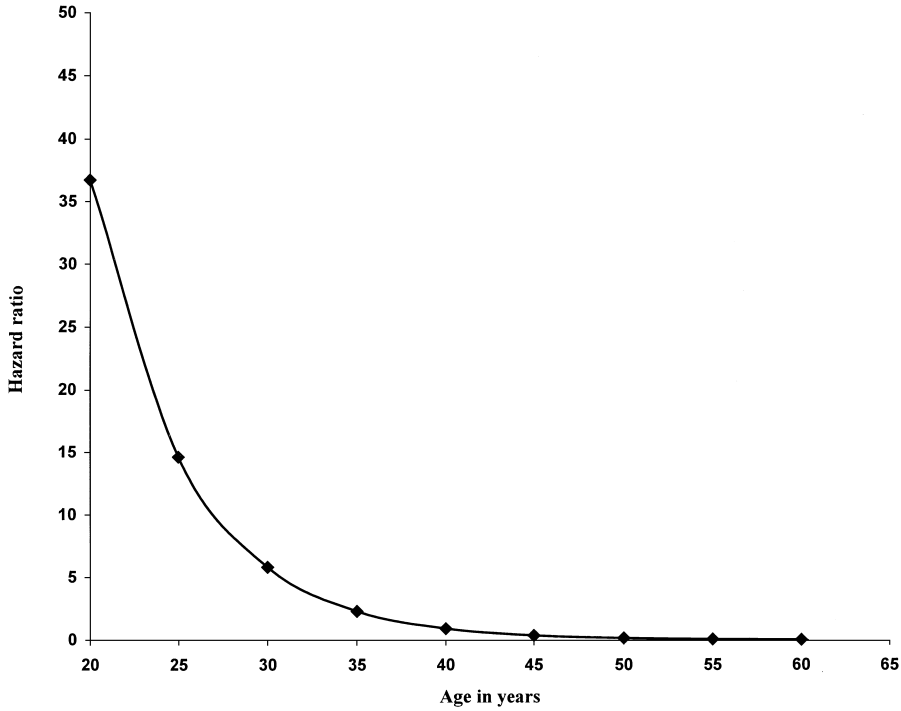

Fig. 2. Estimated hazard ratio with age expressed as a difference from the mean age of pedestrians — undivided street.

expressed as the difference from the mean of these variables. Fig. 2 shows that an averaged age pedestrian has a hazard ratio of one. The figure shows that, since the age coefficient is negative, pedestrians below the mean age have an increased hazard (hazard ratio greater than unity). For pedestrians whose age is above the mean age; the hazard is decreased (less than 1). Stated differently, pedestrians below the average age has greater risk of ceasing their waiting time (i.e. shorter waiting time) at the pedestrian crossing when compared with those whose age is above the mean age. It

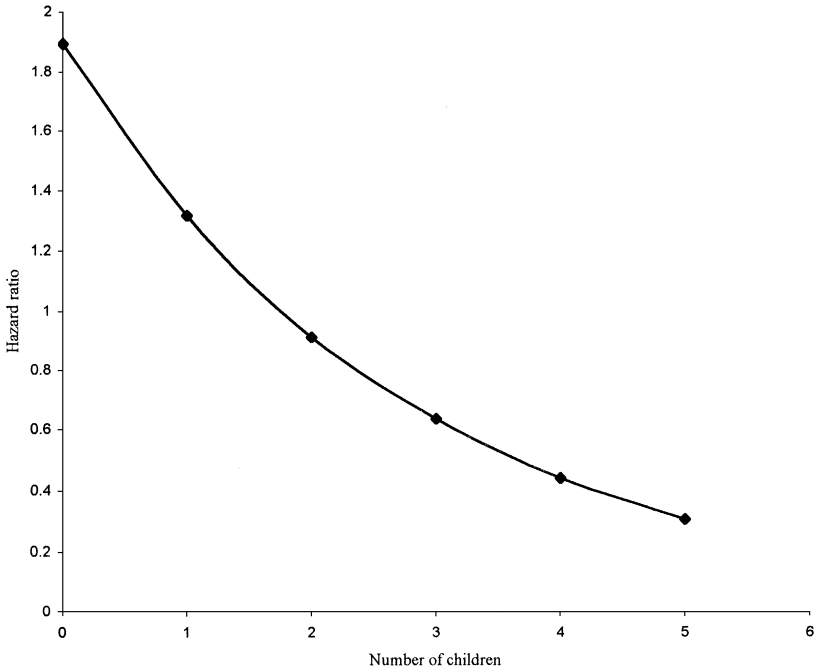

Fig. 3. Estimated hazard ratio with the number of children expressed as a difference from the mean number of children — undivided street.

is interesting to note that as the pedestrian age increases from 20 to 30 years (an increase of 50%) the risk of ending the waiting time at the pedestrian crossing decreases by 84%. These results are in support of similar empirical results reported by Griffiths et al. (1984). Their findings confirmed that the walking speed on pedestrian crossings is different for each age group (young, middle-aged, elderly). As such, each group of pedestrians is likely to have varied perception of dangers when crossing. This perception is based on the pedestrian's expected walking speed.

Table 3 shows the Poisson regression maximum likelihood estimation results for the number of pedestrian's attempts to successfully cross the street. Both the Poisson and the negative binomial regression models were estimated. Estimation results showed that the negative binomial dispersion parameter is not significant (t-ratio = 0.878). The insignificance of the dispersion parameter clearly gives support to the use of the Poisson regression model.

The pedestrian's expected waiting time turned out to have a profound impact on the number of attempts required to cross the street. As the pedestrian's waiting time increases so does the number of attempts required for a successful crossing. This clearly supports the notion that, the longer pedestrians wait the more they attempt to cross the street, i.e. eventually they try to be bold in hopes of forcing approaching vehicles to brake as they cross. However, most of the time they are not successful, as such they continue to make more attempts until they have successfully cross the

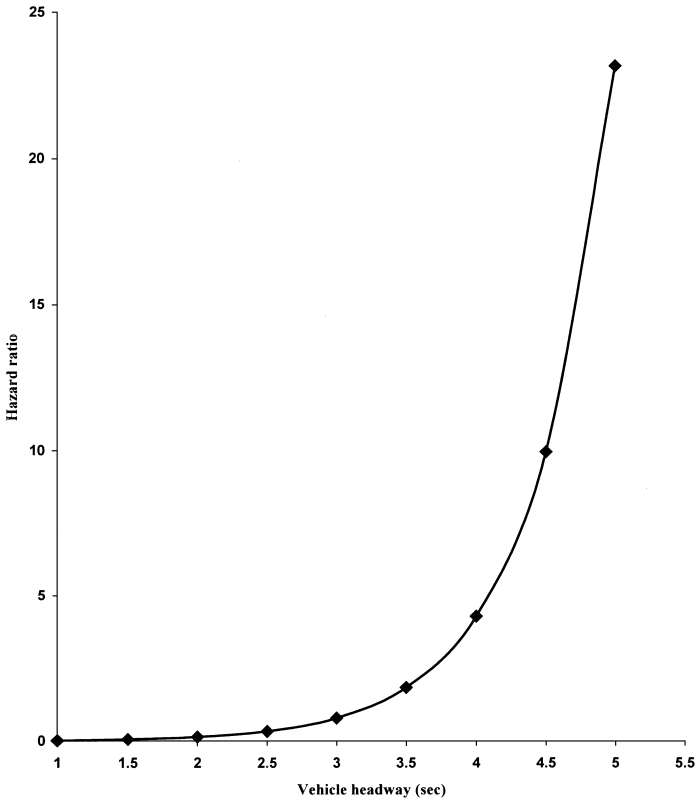

Fig. 4. Estimated hazard ratio with vehicle headway expressed as a difference from the mean vehicle headways — undivided street.

street. Pedestrians who frequently cross pedestrian crossings, come from work and are destined to home, live nearby the pedestrian crossing, single, and are aged between 20 and 30 years make fewer attempts to accomplish a successful crossing. However, pedestrians attempting to accomplish a successful crossing during the morning peak hour (between 7:00 and 8:00) are likely to require a greater amount of attempts. This is expected and reflects the lack of acceptable and safe gaps between vehicles as perceived by pedestrians. This is supported by the positive sign of approaching traffic volume. Griffiths et al. (1984) reported that pedestrians have difficulty in establishing precedence when traffic is heavy and fast moving.

The results in Table 3 also show that the type of approaching vehicle seem to significantly influence the number of attempts needed to cross the street. Pedestrians are likely to cease their waiting and hence number of attempts if the approaching vehicle is a large bus. However, if the approaching vehicle is a passenger vehicle, pedestrians are likely to have greater number of attempts in order to achieve successful crossing.

Table 3
Number of crossing attempts at pedestrian crossings on undivided streets via Poisson[a] regression

Variable	Coefficient/t-ratio[b]
Constant	−6.716/−1.368
Natural Logarithmic of pedestrian waiting time (from Table 1)	0.866/2.434
Crossing frequency (1 if frequently, 0 otherwise)	−1.001/−2.039
Approaching traffic volume (vehicles/min)	−0.003/−1.531
Approaching vehicle speed (1 if between 40 and 60 km/h, 0 otherwise)	−0.391/−1.651
Origin (1 if coming from work and destined to home)	−0.552/−1.774
Living nearby (1 if yes, 0 otherwise)	−0.327/−1.596
Time of day (1 if between 7:00 and 8:00, 0 otherwise)	0.305/2.221
Marital status (1 if single, 0 otherwise)	−0.095/−2.145
Age (1 if between 20 and 30 years, 0 otherwise)	−0.554/−2.200
Type of approaching vehicle (1 if large bus, 0 otherwise)	−0.090/−2.460
Type of approaching vehicle (1 if passenger car, 0 otherwise)	0.153/1.713
Log-likelihood at convergence	−142.385
Chi-squared ($k = 11$)	84.550
Significance level	0.0000

[a] The t-ratio for the negative binomial dispersion parameter turned out to be insignificant (0.878). This result ruled out the adoption of the negative binomial regression model.
[b] Critical values for t-distribution: (1) ±1.645 at the 5% level one-tailed t-test; (2) ±1.282 at the 10% level one-tailed t-test.

4.2. Divided streets

Tables 4 and 5 show the maximum likelihood estimation results for pedestrian's waiting time to cross from one side of the street to refuge and from refuge to the other end respectively. Overall, most parameter estimates are significant and of a plausible sign. In addition to the set of explanatory variables included in the undivided street model (Table 1), one variable relating to the influence of familiarity with the pedestrian crossing was added. Results show that pedestrians who live in the same vicinity of the pedestrian crossing are likely to have higher risk of ceasing the waiting time at the pedestrian crossing. In other words, those pedestrians are likely to have shorter waiting times than those who live far away. The signs of parameter estimates are similar to those reported earlier for undivided streets (Table 1). Table 5 shows two additional explanatory variables that further explain the behavior of pedestrians at pedestrian crossings particularly on divided streets. Estimation results show that as the time required to cross from one side of the street to the refuge island increases so does the risk of ceasing the waiting time from the refuge to the other side of the street. However, the number of attempts made in crossing to the refuge turned out to have no statistical significance over the pedestrian's waiting time to cross from refuge to the other side.

Tables 6 and 7 show estimates of hazard ratios with their corresponding 95% confidence interval for indicator variables for pedestrians crossing from one side of the street to refuge and from refuge to the other side of the street, respectively.

Table 4
Pedestrian's waiting time at pedestrian crossings[a] on divided streets

Variable	Coefficient/t-ratio[b]
If the pedestrian was involved or have witnessed an accident (1 if yes, 0 otherwise)	−6.144/−2.091
If the pedestrian owns and drives a car (1 if yes, 0 otherwise)	−0.478/−1.079
Number of people in the group attempting to cross	−0.159/−0.390
Destined to work (1 if yes, 0 otherwise)	1.081/2.065
Vehicle time headway (in seconds)	6.467/2.096
Number of children in household	−0.232/1.570
Gender (1 if male)	0.302/1.685
Age in years	−0.101/−1.635
Crossing frequency (1 if frequently, 0 otherwise)	0.782/1.991
Living nearby (1 if yes, 0 otherwise)	0.317/1.732
Log-likelihood convergence model	−335.556
Chi-squared ($k = 10$) model	40.247
Significance level model	0.0000
Log-rank test with 9 degrees of:	
Chi-squared	29.000
Probability	0.0012

[a] From one side of the street to central refuge.
[b] Critical values for t-distribution: (1) ±1.645 at the 5% level one-tailed t-test; (2) ±1.282 at the 10% level one-tailed t-test.

Table 5
Pedestrian's waiting time at pedestrian crossings[a] on divided streets

Variable	Coefficient/t-ratio[b]
If the pedestrian was involved or have witnessed an accident (1 if yes, 0 otherwise)	−0.385/−1.410
If the pedestrian owns and drives a car (1 if yes, 0 otherwise)	−0.589/3.026
Number of people attempting to cross	1.031/2.332
Destined to work (1 if yes, 0 otherwise)	1.376/2.886
Vehicle time headway (in seconds)	–
Number of children in household	−0.102/−1.609
Gender (1 if male)	3.133/3.164
Age in years	−0.097/−1.591
Crossing frequency (1 if frequently, 0 otherwise)	0.349/1.424
Living nearby (1 if yes, 0 otherwise)	4.938/4.492
Expected pedestrian's waiting time to central refuge island	0.307/4.331
Expected number of attempts to central refuge island	0.045/0.198
Log-likelihood convergence model	−337.522
Chi-squared ($k = 12$) model	61.789
Significance level model	0.0000
Log-rank test with 9 degrees of:	
Chi-squared	66.632
Probability	0.0000

[a] From central refuge to the other side of the street.
[b] Critical values for t-distribution: (1) ±1.645 at the 5% level one-tailed t-test; (2) ±1.282 at the 10% level one-tailed t-test.

Table 6
Estimates of hazard ratios[a] for pedestrians crossing divided streets[b]

Indicator variable	Hazard ratio	95% Confidence interval
If pedestrian was not involved or did not witness a pedestrian accident	1.155	0.933–1.405
If a pedestrian has no access to a private car	1.613	0.952–5.961
If a pedestrian is destined to work	2.948	1.058–8.216
Gender (1 if male)	1.353	0.952–1.921
Crossing frequency (1 if frequently)	2.186	1.014–4.713
Living nearby (1 if yes)	1.373	0.959–1.965

[a] For indicator variables.
[b] From one side of the street to central refuge.

Table 7
Estimates of hazard ratios[a] for pedestrians crossing divided streets[b]

Indicator variable	Hazard ratio	95% Confidence interval
If pedestrian was not involved or did not witness a pedestrian accident	1.462	0.932–3.081
If a pedestrian has no access to a private car	1.802	0.970–2.215
If a pedestrian is destined to work	3.959	1.554–10.084
Gender (if male)	3.105	1.539–6.263
Crossing frequency (1 if frequently)	1.418	0.877–2.292
Living nearby (1 if yes)	2.555	1.696–3.848

[a] For indicator variables.
[b] From central refuge to the other side of the street.

Estimates seem to suggest the different waiting times pedestrians have as they cross from one side of the street to refuge and from refuge to the other side of the street influence pedestrians' behavior. For example, male pedestrians are 1.35 times more likely to have shorter waiting times when crossing from one side to the refuge than female pedestrians. However, males are 3.105 times more likely to have shorter waiting time in crossing from the refuge to the other side of the street. In addition, pedestrians who live in the vicinity of the pedestrian crossing are 1.37 more likely to have shorter waiting times when crossing from one side of the street to refuge. However, they are 2.56 times more likely to have a shorter waiting time in crossing from the refuge to the other side of the street. Similarly pedestrians destined to work are 2.95 times more likely to have shorter waiting time when crossing to refuge. However, they are 3.96 times more likely to have shorter waiting times when crossing from refuge to the other side of the street.

Tables 8 and 9 show the maximum likelihood estimation results for the number of crossing attempts needed to cross from one side of the street to refuge and from refuge to the other side respectively. Both the Poisson and the negative binomial regression models were estimated for each direction of divided streets. Negative binomial regression results revealed that the dispersion parameter was insignificant

Table 8
Number of crossing[a] attempts on divided streets via Poisson[b] regression

Variable	Coefficient/t-ratio[c]
Constant	2.384/1.017
Expected pedestrian's waiting time	0.015/2.334
Crossing frequency (1 if frequently, 0 otherwise)	−0.520/−1.700
Approaching traffic volume (vehicles/min)	0.039/2.217
Approaching vehicle speed (1 if between 40 and 60 km/h, 0 otherwise)	−0.433/−1.900
Origin (1 if coming from work)	−0.849/−2.145
Living nearby (1 if yes, 0 otherwise)	−0.121/−1.499
Time of day (1 if between 10:00 am and 11:00 am, 0 otherwise)	−0.568/−1.454
Marital status (1 if single, 0 otherwise)	−0.526/−1.301
Age (1 if between 20 and 30 years, 0 otherwise)	−0.171/−1.358
Type of approaching vehicle (1 if large bus, 0 otherwise)	−0.054/−1.654
Type of approaching vehicle (1 if passenger car, 0 otherwise)	0.027/1.671
Log-likelihood at convergence	−118.469
Chi-squared ($k = 11$)	45.701
Significance level	0.0000

[a] From one side of the street to central refuge.
[b] The t-ratio for the negative binomial dispersion parameter turned out to be insignificant (0.712). This result ruled out the adoption of the negative binomial regression model.
[c] Critical values for t-distribution: (1) ±1.645 at the 5% level one-tailed t-test; (2) ±1.282 at the 10% level one-tailed t-test.

for pedestrians crossing from one side of the street to refuge (t-ratio = 0.712) and for pedestrians crossing from refuge to the other side of the street (t-ratio = 0.480). These results underscore the appropriateness of the estimated Poisson regression models.

Again, the waiting time turned out to profoundly impact the number of crossing attempts. As the pedestrian's waiting time increases so does the number of attempts required for a successful cross (Tables 8 and 9). In addition, the results show the significance of the type of approaching vehicle on the number of attempts. Similar to what was reported in Table 3, pedestrians are likely to cease their waiting time and thus reduce the number of attempts if the approaching vehicle was a large bus. In addition to the set of variables stated in Table 8, number of crossing attempts variable was included in Table 9. The results seem to indicate that the number of crossing attempts made to cross from one side of the street to central refuge negatively influence the number of crossing attempts from refuge to the other side of the street.

5. Summary and conclusion

This paper introduces a methodology, combining duration and count models to study the behavior of pedestrians at pedestrian crossings located on divided and

Table 9
Number of crossing[a] attempts on divided streets via Poisson[b] regression

Variable	Coefficient/t-value[c]
Constant	−10.83/−2.785
Expected pedestrian's waiting time to central refuge	0.243/3.070
Crossing frequency (1 if frequently, 0 otherwise)	0–0.640/−1.586
Approaching traffic volume (vehicles/min)	0.001/1.840
Approaching vehicle speed (1 if between 40 and 60 km/h, 0 otherwise)	1.647/2.046
Origin (1 if coming from work, 0 otherwise)	−0.705/−1.490
Living nearby (1 if yes, 0 otherwise)	0.216/0.720
Time of day (1 if between 10:00 am and 11:00 am, 0 otherwise)	0.167/1.418
Marital status (1 if married, 0 otherwise)	0.427/1.326
Age (1 if over 40 years, 0 otherwise)	0.051/1.668
Type of approaching vehicle (1 if large bus, 0 otherwise)	−0.110/−2.191
Type of approaching vehicle (1 if passenger car, 0 otherwise)	0.071/1.501
Expected number of crossing attempts to central refuge	−0.091/−1.837
Log-likelihood at convergence	−106.687
Chi-squared ($k = 12$)	42.371
Significance level	0.0000

[a] From central refuge to the other side of street.
[b] The t-ratio for the negative binomial dispersion parameter turned out to be insignificant (0.480). This result ruled out the adoption of the negative binomial regression model.
[c] Critical values for t-distribution: (1) ±1.645 at the 5% level one-tailed t-test; (2) ±1.282 at the 10% level one-tailed t-test.

undivided urban streets. Findings from this paper are partly supplemented by those found in previous research that is based on other theoretical derived models, in particular, those with a specifically attitudinal focus.

Proportional hazard models are specified and estimated to identify the determinants of pedestrians waiting time on pedestrian crossings. In addition, Poisson regression models are specified and estimated to model the number of attempts required for successful crossing. Estimated Poisson regression models clearly indicate the inappropriateness of the negative binomial regression structure to model number of attempts needed to cross the street. The pedestrian's expected waiting time was included in the number of attempts model. Pedestrian's behavior at divided streets were modeled by estimating separate models for the pedestrian's waiting time and number of crossing attempts from one side of the street to a central refuge island and from refuge to the other side of the street.

Estimation results for divided and undivided streets suggested predictors that are likely to make pedestrians accept higher risk and cease their waiting time and other predictors that are likely to lower the risk and extend their waiting time at pedestrian crossings. For example, pedestrians who frequently use a certain pedestrian crossing and who live nearby the crossing are likely to accept higher risk and cease their waiting time at the pedestrian crossing. On the other hand, the pedestrian's past involvement in a traffic accident seem to inhibit the pedestrian from accepting higher risk to cease their waiting time (i.e. they have higher waiting time than those

who had no previous involvement). Furthermore, pedestrians who have access to private vehicles seem to be more aware of the risk involved. Therefore, they are more cautious of time needed before crossing. In addition, results reported in this paper clearly indicated that the pedestrian's expected waiting time has a profound impact on the number of attempts required by the pedestrian to make a successful crossing.

Estimated duration and count models for divided streets show some interesting results. First, pedestrians who spend more time waiting to cross from one side of the street to central refuge are likely to have a higher risk of ending their waiting time as they cross from the central refuge to the other side of the street (i.e. the waiting time from refuge to the other side of street decreases). However, the number of attempts made to cross from one side to central refuge from turned out to have no statistical significance over the pedestrian's waiting time from the central refuge to the other side of the street. It seems pedestrians are more sensitive to the waiting time than number of attempts. Secondly, estimates of hazard ratios suggest that pedestrians seem to behave differently as they cross from one side of the street to central refuge and from central refuge to the other side of the street. This is manifested by the waiting time that the pedestrian is able to tolerate. For example, male pedestrians are 1.35 times more likely to have shorter waiting times to cross from one side to refuge than female pedestrians. However, males are 3.105 times more likely to have shorter waiting times to cross from refuge to the other side of the street.

There are policy implications that follow from some of the empirical results obtained. Firstly, the results indicate that approaching traffic volume and vehicle speeds are instrumental in determining the pedestrian's waiting time (delay) and the number of crossing attempts. Thus, punishment programs directed towards drivers by traffic law enforcement are likely to be instrumental in suppressing undesired risky behavior. Measures such as small fines for offending drivers are not likely to be good deterrent for offences (Groeger and Rothengatter, 1998). Repeated offenders who violate pedestrian crossings should be subject to having their driver license revoked. In addition, these drivers should attend specially designed pedestrian-oriented compulsory training courses. Secondly, pedestrians should be targeted through educational and public information awareness campaigns to deter pedestrians from engaging in risky behavior and to promote safe road crossing behavior. Such programs could be applied through mass media. Groeger and Rothengatter (1998) reported that educational programs for road crossing skills proved very effective. It is hoped that this piece of work will increase the sophistication of measurement in this area to better understand pedestrian's behavior at pedestrian crossings.

References

Andersen, P.K., 1991. Survival analysis 1982–1991: the second decade of the proportional hazards regression model. Statistics in Medicine 10, 1931–1941.

Chap, T.L., 1997. Applied Survival Analysis. John Wiley, New York.

Clark, S.J., Perry, J.N., 1989. Estimation of the negative binomial parameter k by maximum quasi-likelihood. Biometrics 45, 309–316.

Cox, D.R., 1972. Regression models and life-tables. Journal of the Royal Statistical Society B34, 187–220.

Cresswell, C., Hunt, J.G., 1979. Site evaluation of a zebra crossing simulation model. Traffic Engineering and Control 20 (10), 467–470.

Fegan, J.C., 1978. Major engineering approaches toward increasing pedestrian safety. Public Roads 42 (3), 85–90.

Gourieroux, C., Montfort, A., Trognon, A., 1984. Pseudo-maximum likelihood methods: applications to Poisson models. Econometrica 52, 701–720.

Griffiths, J.D., Hunt, J.G., Marlow, M., 1984. Delays at pedestrian crossings. Traffic Engineering and Control 25, 365–371.

Griffiths, J.D., Hunt, J.G., Cresswell, C., 1976. Warrants for zebra and pelican crossings using a minimum waiting cost criterion. Traffic Engineering and Control 17, 59–62.

Groeger, J.A., Rothengatter, J.A., 1998. Traffic psychology and behaviour. Transportation Research part F1 (1), 1–9.

Heckman, J.J., Singer, B., 1984. A method of minimizing the impact of distributional assumptions in econometric models for duration data. Econometrica 52, 271–320.

Hensher, D.A., Mannering, F.L., 1994. Hazard-based duration models and their application to transport analysis. Transport Review 14 (1), 63–82.

Himanen, V., Kulmala, R., 1988. An application of logit models in analyzing the behavior of pedestrians and car drivers on pedestrians crossings. Accident Analysis and Prevention 2 (3), 187–197.

Jovanis, P.P., Chang, H.-L., 1989. Disaggregate model of highway accident occurrence using survival theory. Accident Analysis and Prevention 21 (5), 445–458.

Katz, A., Zaidel, D., Elgrishi, A., 1975. An experimental study of driver and pedestrian interaction during the crossing conflict. Human Factors 17 (5), 514–527.

Landles, J.R., 1983. The overall effect on accidents at sites where zebra crossings were installed. Traffic Engineering and Control 24, 9–11.

Lawless, J.F., 1987. Negative binomial and mixed Poisson regression. The Canadian Journal of Statistics 15 (3), 209–225.

Paselk, T.A., Mannering, F.L., 1994. Use of duration models for predicting vehicular delay at a US/Canadian border crossing. Transportation 21 (3), 249–270.

Song, L., Black, J., Dunne, M., 1993. Environmental capacity based on pedestrian delay and accident risk. Road and Transport Research 2 (3), 40–50.

Varhelyi, A., 1998. Drivers' speed behavior at a zebra crossing: a case study. Accident Analysis and Prevention 30 (6), 731–743.